PHYSICAL SCIENCES

PHYSICAL SCIENCES GRADE 12 TERM 4 CONTROLLED ASSESSMENT PACK

SUBJECT	PHYSICAL SCIENCES
GRADE	12
ASSESSMENT TASK	PHYSICAL SCIENCES GRADE 12 TERM 4 CONTROLLED ASSESSMENT PACK
MARKS	50
TIME	1 hour

Educator approval required before classroom use.

INSTRUCTIONS AND INFORMATION

1. Read all instructions carefully.

2. Answer all questions or complete all activities as instructed.

3. Show all working where calculations, planning, diagrams or explanations are required.

4. Use the space provided by the educator, answer booklet or learner task book.

5. Write neatly and clearly.

SOURCE MATERIAL

SOURCE 1: Official-paper exemplar: Data Table

1.7 The table below shows the ionisation constants, K_a, for two acids at 25 °C.

ACID	K_a
Butanoic acid	$1,5 \times 10^{-5}$
Ethanoic acid	$1,8 \times 10^{-5}$

Consider the following statements for these two acids when they have equal concentration at 25 °C:

(i) Both are weak acids.
(ii) Butanoic acid is a stronger acid than ethanoic acid.
(iii) The butanoic acid solution has a lower concentration of hydronium ion, $H_3O^+(aq)$, than the ethanoic acid solution.

Which of the above statements are TRUE?

A (i) and (ii) only

B (i) and (iii) only

C (ii) and (iii) only

D (i), (ii) and (iii) (2)

1.8 Which ONE of the following pairs of acids and bases, all of the same concentration, react to give the highest pH at the equivalence point in a titration at 25 °C?

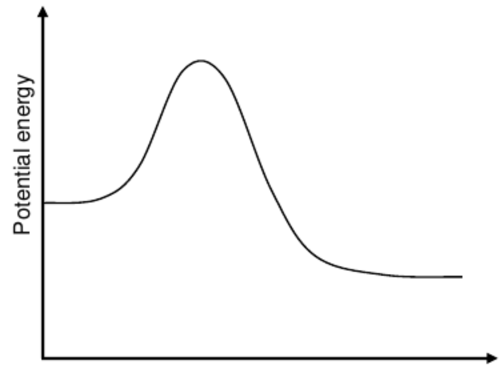
A $HC\ell$ and NH_3

B $HC\ell$ and NaOH

SOURCE 2: Official-paper exemplar: Diagram Or Model

1.4 The potential energy diagram below is for the following hypothetical chemical reaction:

$$X + Y \rightleftharpoons Z$$



Which ONE of the following combinations of values for the heat of the reaction and the activation energies can be obtained for this reaction?

	$\Delta H_{(forward)}$ ($kJ \cdot mol^{-1}$)	**$E_{A(forward)}$ ($kJ \cdot mol^{-1}$)**	**$E_{A(reverse)}$ ($kJ \cdot mol^{-1}$)**
A	-400	300	100
B	-200	300	100
C	+400	100	300
D	-200	100	300

(2)

1.5 Initially, an equal number of moles of hydrogen gas, $H_2(g)$, and iodine gas, $I_2(g)$, are mixed in a closed container. The reaction reaches equilibrium at a

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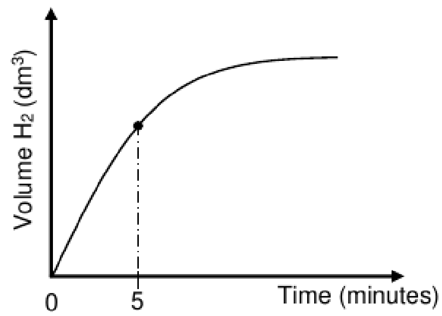
SOURCE 3: Official-paper exemplar: Graph Or Chart

The graph of volume $H_2(g)$ versus time for this experiment, not drawn to scale, is shown below.



5.1.2 For the time interval t = 0 to t = 5 minutes, the average reaction rate for the formation of $H_2(g)$ is 0,033 $dm^3 \cdot min^{-1}$.

Calculate the mass of Aℓ present in the container at t = 5 minutes. Take the molar gas volume as 24,5 $dm^3 \cdot mol^{-1}$. (6)

Assume that the concentration of the HCℓ(aq) stays constant for the duration of the reaction.

5.1.3 Use the collision theory to explain the change in the reaction rate from t = 0 to t = 5 minutes. (4)

EXPERIMENT II

Experiment I is repeated using a 2 $mol \cdot dm^{-3}$ HCℓ solution.

5.1.4 Redraw the above graph (NO numerical values need to be shown)

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QUESTIONS

QUESTION 1: SECTION A: CONCEPTS AND DATA

Instructions: Use the supplied data, diagram or graph.

Stimulus: Data/diagram stimulus linked to final integrated Physical Sciences examination practice.

1.1. Define one key concept from the term focus. **(2)**

1.2. Describe the trend or relationship shown by the supplied data. **(5)**

1.3. Explain the process or principle using correct terminology. **(8)**

1.4. Calculate or infer a result from the supplied values and show working where needed. **(5)**

QUESTION 2: SECTION B: INVESTIGATION DESIGN

Instructions: Critique method, variables and evidence.

Stimulus: Investigation focus: mixed calculations, graph interpretation and experimental design.

2.1. State a suitable aim or hypothesis for the investigation. **(3)**

2.2. Identify independent, dependent and controlled variables. **(6)**

2.3. Evaluate validity and reliability and suggest one improvement. **(6)**

QUESTION 3: SECTION C: APPLICATION

Instructions: Use subject knowledge and supplied evidence.

Stimulus: Application focus: source-linked final revision paper with memo-ready mark allocation.

3.1. Answer an integrated explanation question using evidence from the stimulus. **(15)**